Infant Sleep Training: Graduated Extinction & The Ferber Method

RCT and cohort evidence finds no measured attachment or behavioral harm from behavioral sleep training, graduated extinction (the Ferber method), and cry-it-out through age 6.

GRADE: LOW-TO-MODERATE | STATUS: DRAFT | DOMAIN: REGULATION, HEALTH, PARENTING | REPLICATION: VERIFIED

WHAT WE KNOW (Narrowest Supported Conclusion)

Randomized and longitudinal evidence does not find that graduated extinction (the Ferber method), unmodified extinction (Marc Weissbluth / cry it out / CIO), bedtime fading, or related behavioral infant sleep training (applied across nighttime sleep, daytime naps, crib transitions, and pacifier routines, including commercial schedules such as Moms on Call, Taking Cara Babies, Precious Little Sleep, Babywise / Bucknam & Ezzo, Suzy Giordano, Harvey Karp, Craig Canapari, Tracy Hogg, Gina Ford, Penelope Leach, and James McKenna) harms measured infant–parent attachment or increases behavioral/emotional problems through at least age 6, though most trials are underpowered for rare or long-latency harms.

EVIDENCE AND CAUSAL ASSESSMENT

- Price, Wake, Ukoumunne & Hiscock (2012), Pediatrics: five-year follow-up of a randomized infant sleep intervention found no differences in child sleep, mental health, or child–parent relationship at age 6. The previously listed Sleep Health (2017) paper by those four authors was not located.

- Gradisar et al. (2016), Pediatrics RCT (n=43): graduated extinction and bedtime fading improved some sleep measures versus education control. At 12 months there were no significant group differences in Strange Situation attachment or emotional/behavioral problems; salivary cortisol did not show an adverse stress signal. This is the 2016 Pediatrics paper. It is not a 52-study meta-analysis.

- Bilgin & Wolke (2020), JCPP: observational cohort of 178 dyads. Maternal report of leaving the infant to cry it out in the first six months was not associated with insecure or disorganized attachment at 18 months. This is not an RCT of a sleep-training protocol.

- Mindell, Kuhn, Lewin, Meltzer & Sadeh (2006), SLEEP: AASM review of 52 treatment studies (n>2,500), nine of them Level I RCTs. Behavioral interventions were generally reported as efficacious for bedtime problems. This is an efficacy synthesis, not a 2016 Pediatrics harm meta-analysis. The retired catalog ID hiscock-2016 identified none of these papers.

- Williamson, Mindell, Hiscock & Quach (2019) describes LSAC sleep-problem trajectories. It is not a sleep-training RCT and does not license claims about extinction methods, cortisol, or attachment.

EFFECT SIZE AND PERSISTENCE

Reported effects are null for attachment security and behavior problems in the RCTs and follow-ups located here, tracked out to roughly age 6 in the longest follow-up. Sleep improvements in Gradisar 2016 and in Mindell 2006's efficacy review are a different claim from harm.

BIAS, HETEROGENEITY, AND CONFOUNDING

GI attenuation: n/a (randomized behavioral intervention trials and follow-ups, not an observational phenotype).

Most trials are relatively small to moderate and rely heavily on parent-report sleep measures; Gradisar's attachment comparison is underpowered. Self-selection into sleep-training studies may limit generalizability. Bilgin & Wolke is observational. Longer-term outcomes and effects in infants with heightened biological vulnerability are not well covered.

WHAT THIS DOES NOT MEAN (Strict Boundary Conditions)

Does not establish that sleep training is necessary or that not sleep-training risks poorer outcomes; absence of demonstrated harm is not equivalent to a demonstrated universal safety guarantee at the individual level.

WHAT ACTION IS REASONABLY JUSTIFIED

Families choosing structured sleep-training methods are not acting against the weight of current evidence on attachment or behavioral harm; this is a values/preference-consistent decision domain more than an evidence-mandated one.

VERIFIED SOURCES & DOI IDENTIFIERS

- Price, Wake, Ukoumunne & Hiscock (2012), Pediatrics, five-year follow-up of the Melbourne infant sleep RCT

- Gradisar et al. (2016), Pediatrics RCT (DOI 10.1542/peds.2015-1486)
- Bilgin & Wolke (2020), JCPP (DOI 10.1111/jcpp.13223)
- Mindell et al. (2006), SLEEP AASM review (DOI 10.1093/sleep/29.10.1263)
- Williamson, Mindell, Hiscock & Quach (2019), LSAC sleep-problem trajectories (not a sleep-training RCT)